Hamsavardhini S

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EDUCATION

M.Sc. Data Science (Integrated) Expected Graduation: May 2028
PSG College of Technology, Coimbatore, Tamil Nadu CGPA: 7.89/10
(**Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management System, Linear Algebra, Probability & Statistics, Stochastic Processes, Optimization Techniques, Machine Learning, Deep Learning)

XII (State board) 2023
Krishnasamy Memorial Matriculation Hr. Sec. School, Cuddalore, Tamil Nadu Percentage: 93%

TECHNICAL SKILLS

Languages: Python, R, C++, C
ML & DS Frameworks: PyTorch, TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas, SciPy, Statsmodels
Databases & Backend: MySQL, MongoDB, Neo4j, FastAPI
Visualization & Analytics: Power BI, Excel, Matplotlib, Seaborn
Cloud & Developer Platforms: AWS (EC2, S3), Docker, Git, Linux

EXPERIENCE

Research Intern | IIIT Bangalore *Jun 2026 – Present*

- Contributing to research on Quantitative Analysis of Fetal Brain Development from 3D ultrasound volumes at the E-Health Research Center (EHRC), IIITB, in collaboration with ICMR and NIMHANS.
- Currently building a robust pipeline integrating anatomical orientation normalization, geometric refinement, volume registration, and deep learning-based segmentation to enable reliable quantitative analysis across gestational ages.

Machine Learning Intern | Exa Protocol *Mar 2025 – Sep 2025*

- Engineered end-to-end Retrieval-Augmented Generation (RAG) pipeline and fine-tuned Large Language Models (LLMs) using LoRA/QLoRA to enhance domain adaptation and task-specific performance.
- Designed multi-agent AI systems by experimenting with various agent frameworks and MCP-based architectures for intelligent task orchestration, context management, and agent collaboration.

PROJECTS

Alpha Factor Research & Portfolio Construction Engine

- Built an end-to-end US/India equity pipeline where validated factors (Newey-West testing, cross-market transfer checks) feed a linear/ML signal, sized into a risk-gated trade list that rejects and re-optimizes on limit breaches.
- Engineered production rigor with purged cross-validation to prevent ML leakage, an event-driven backtester with realistic costs, and live monitoring with risk alerts packaged as a containerized, CI-tested dashboard.

Bayesian Network Based Predictive Maintenance

- Developed a predictive analytics pipeline for engine health monitoring using multivariate time-series sensor data, feature engineering, PCA-based health scoring, and Hidden Markov Models (HMMs) to model system degradation.
- Developed a Bayesian Network-based probabilistic model to estimate failure probability and predict Remaining Useful Life (RUL), enabling data-driven maintenance planning and improved engine reliability.

Adaptive Multi-Agent Logistics Optimization System

- Designed a multi-agent logistics optimization system with an MCP-style architecture, implementing parallel agent orchestration and route planning algorithms for real time vehicle routing and adaptive decision making.
- Built intelligent agents for constraint handling, conflict resolution, and trigger based replanning, enabling dynamic rerouting under changing conditions with continuous SLA compliance monitoring.

ACHIEVEMENTS & ACTIVITIES

- Selected participant of the ACM India Summer School on IoT & Next-Generation Networks (2025)
- Placement Representative for the 2023–28 Data Science batch at PSG Tech, coordinating placement activities.
- Active member, Entrepreneurship Club at PSGCT, contributing to planning and execution of events.